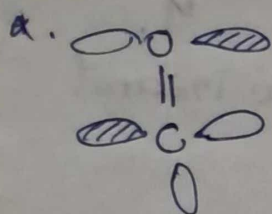
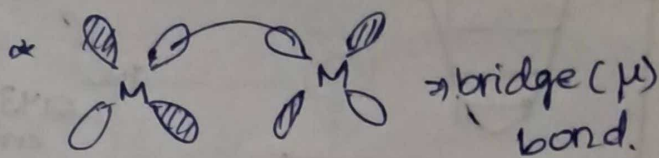
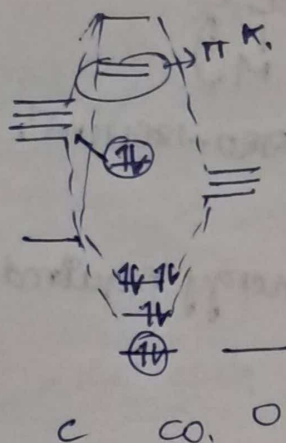


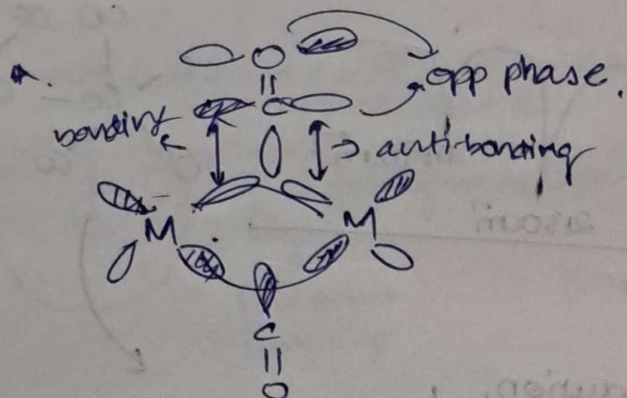
Organometallic complexes.



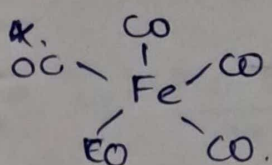
co-soft base.



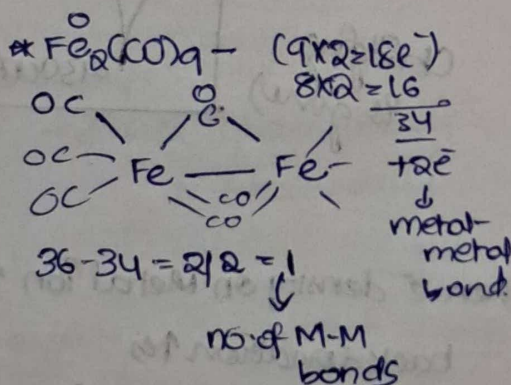
bridge (μ) bond.



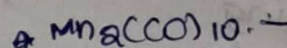
bonding $\Rightarrow$ anti-bonding $\Rightarrow$ half bonding + half anti bonding $\Rightarrow$ nonbonding



Fe(CO)_5 .

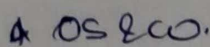


$$\begin{aligned} & \text{Fe}_2(\text{CO})_9 - (9 \times 2 = 18e^-) \\ & 8 \times 2 = 16 \\ & \underline{34} \\ & + 2e^- \\ & \downarrow \\ & \text{metal-metal bond} \\ & 36 - 34 = 2/2 = 1 \\ & \downarrow \\ & \text{no. of M-M bonds} \end{aligned}$$

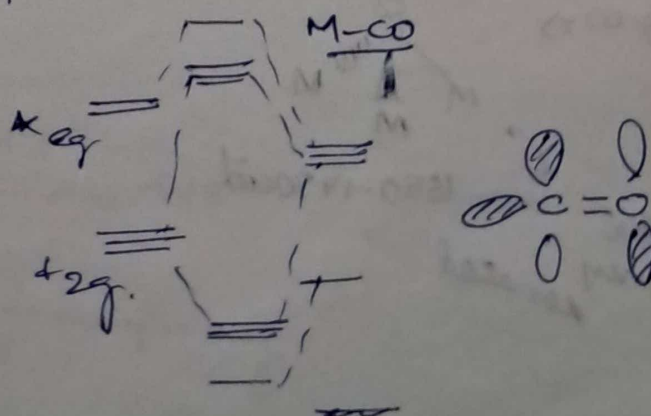
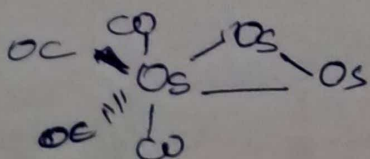


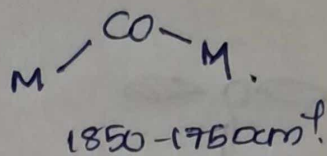
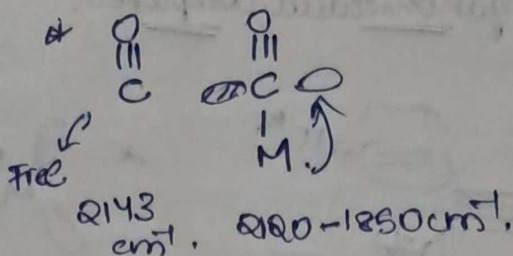
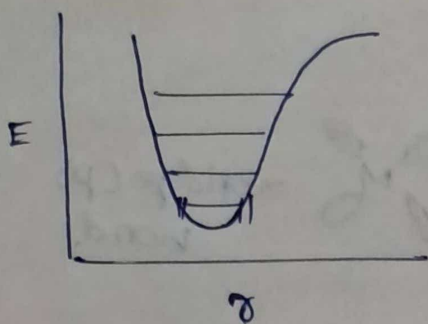
$$\begin{aligned} 7 \times 2 &= 14 \\ 10 \times 2 &= 20 \\ \underline{34} \end{aligned}$$

$$36 - 34 = 2/2 = 1$$

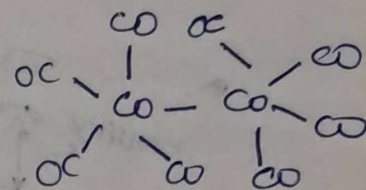
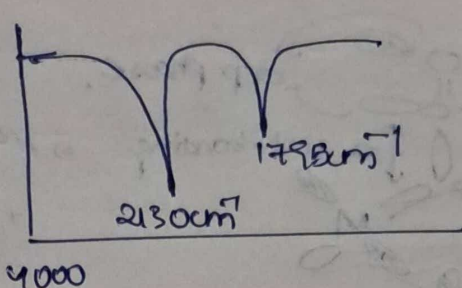
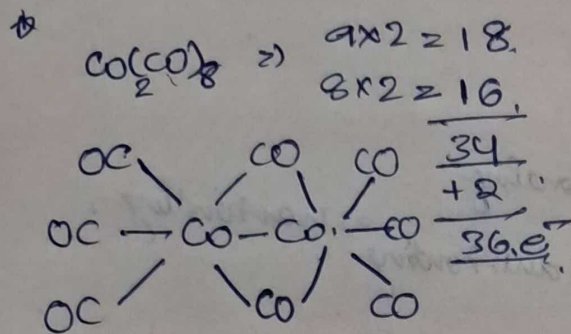


$\rightarrow$ 3rd period $\Rightarrow$ no comparable energy l.w Os & Co
 $\Rightarrow$ no bridging.





the higher the bond order, higher the energy required to excite it.

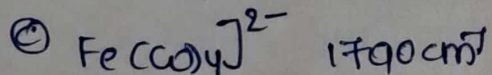
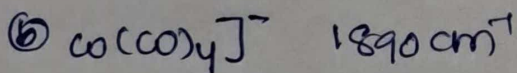
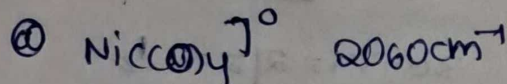
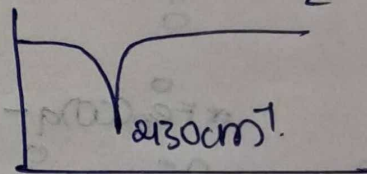


2 types of IR spectrum

- 1) bridging
- 2) terminal.

In solution,

bridging is gone

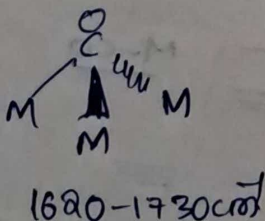


when e^- density on Metal ion $\uparrow$ es,

backdonation $\uparrow$ es

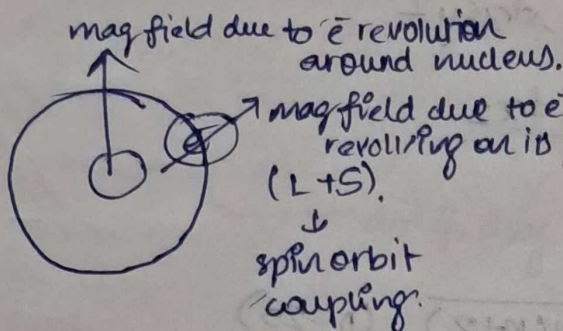
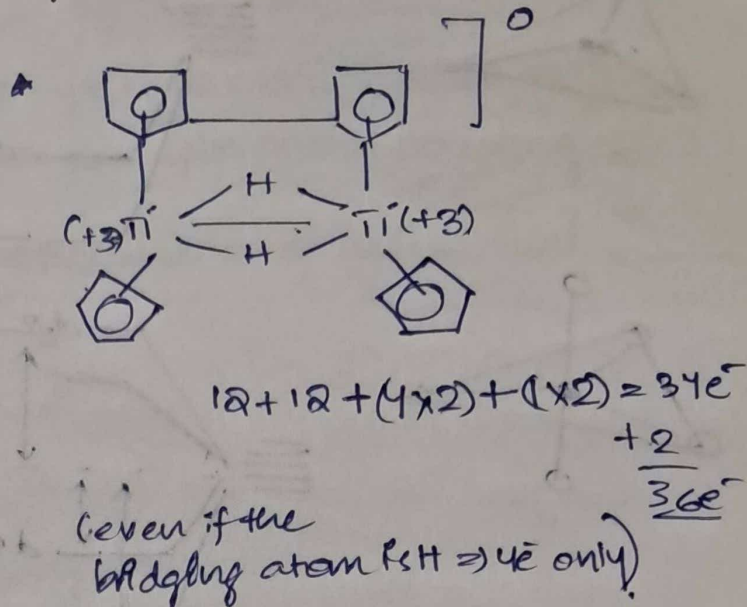
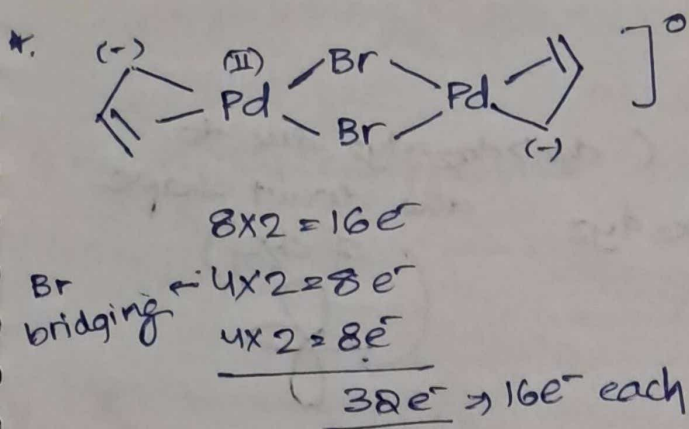
amount of e^- in antibonding MO $\uparrow$ es
bond order $\downarrow$ es.

* $\mu_3\text{-CO} \Rightarrow$



2e only donated

* Crystal field theory and Ligand field theory:



$$\mu_{SO} = \sqrt{n(n+2)} \mu_B$$

(if no. of unpaired e^- is more, conductivity generally increases.
 sometimes decreases)

metal complex geometry hybridization

$\rightarrow$ largest angles.
 $\rightarrow$ either square planar or tetrahedral

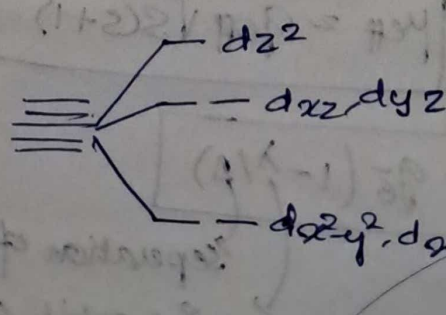
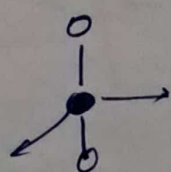
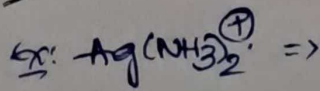
$\tau_4 = \frac{360 - (\alpha + \beta)}{142}$
 $\tau_4 = 0 \rightarrow$ sq planar
 $\tau_4 = 1 \rightarrow$ tetrahedral

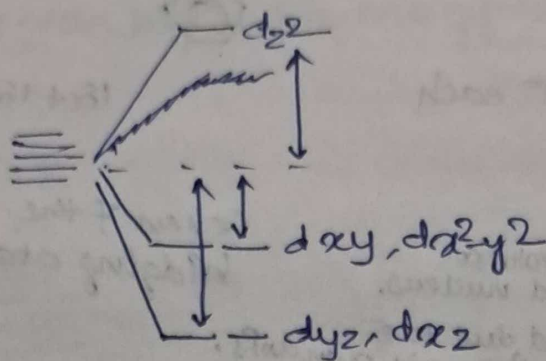
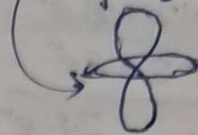
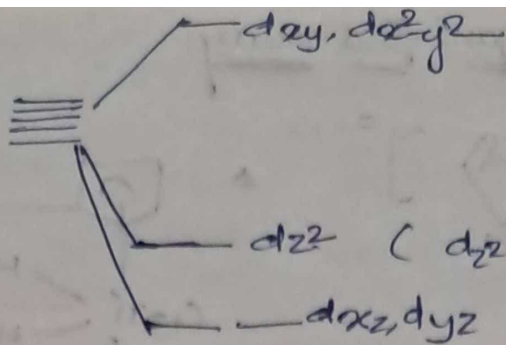
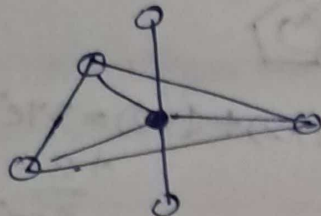
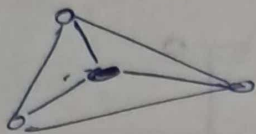
For 5 coordinate system, $\tau_5 = \frac{360 - (\alpha + \beta)}{60}$
 $\tau_5 = 0 \rightarrow$ sq py
 $\tau_5 = 1 \rightarrow$ tbp.

dsp^3 sp^3d^2
 more reactive

hybrid does not explain the observed stability for many compounds

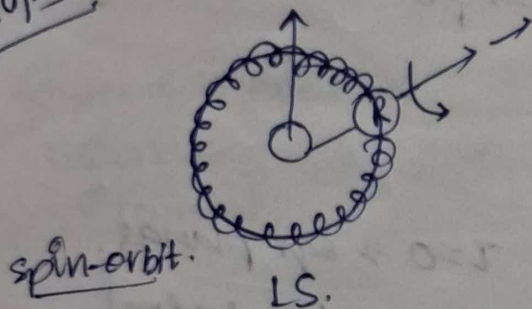
CFT





the energy order remains same but the energy levels always change w.r.t the ligand nature (strength).

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$$\mu_{SO} = \sqrt{n(n+2)} \quad (n=2S)$$

$$= \sqrt{2S(2S+2)}$$

$$= 2\sqrt{S(S+1)}$$

$$L = 2S \times \frac{1}{2}$$

Lande-splitting factor.

$$= g$$

g gyromagnetic ratio.

angular momentum in orbit

g for free $e^- = 2$. (cause its not localised)

g value is different for each orbital.

$$\mu_{SO} \rightarrow \chi \quad \mu_{eff} \checkmark$$

$$\Rightarrow \mu_{eff} = g_{eff}(1 - \lambda/\Delta)$$

$$\Rightarrow \boxed{\mu_{eff} = g_{eff} \sqrt{S(S+1)}}$$

$$\boxed{g_{eff} = g_e (1 - \lambda/\Delta)}$$

separation of d-orbitals
spin orbit coupling constant.

$\lambda \Rightarrow$ if it is less than half filled $\Rightarrow +ve$; more than half filled $\Rightarrow -ve$

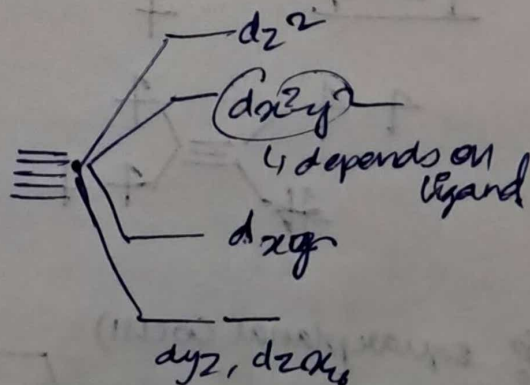
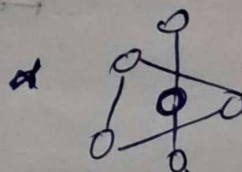
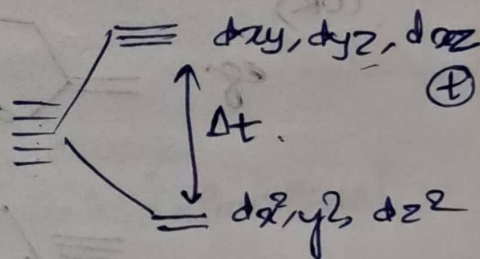
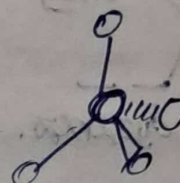
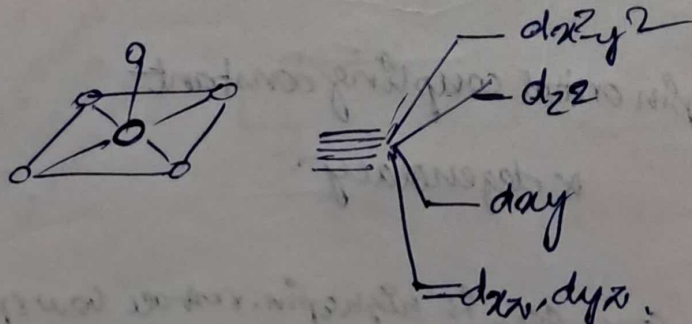
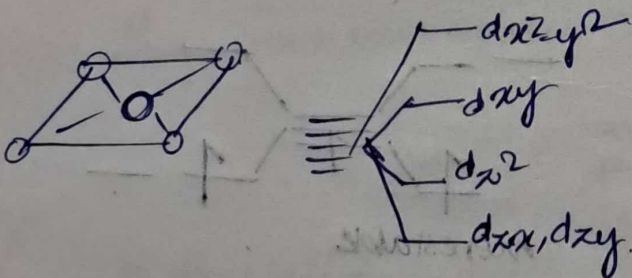
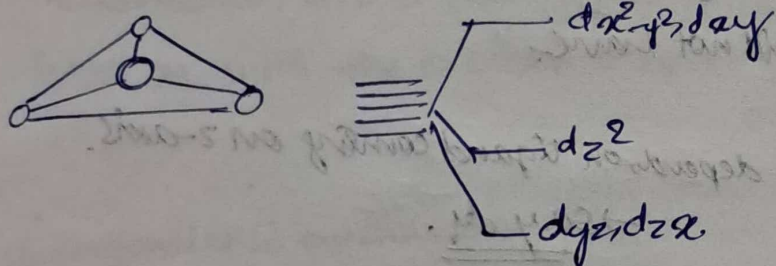
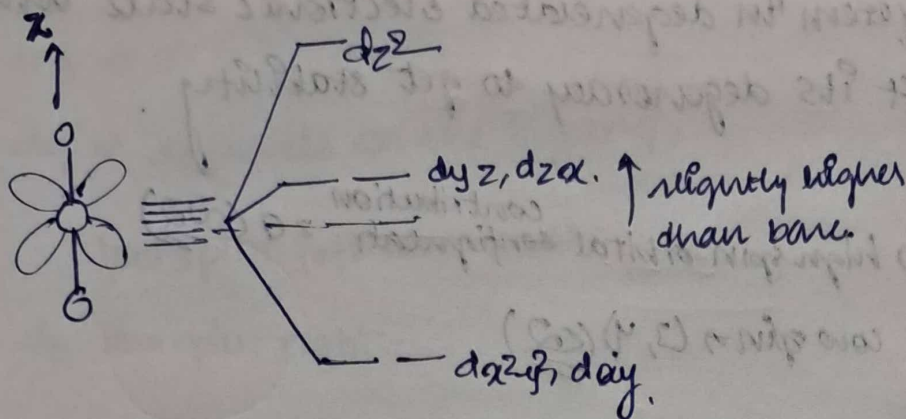
Ex: $Cr^{+3} \lambda +ve. \Rightarrow g_{eff} < 2$
 $Cu^{+2} \lambda -ve. \Rightarrow g_{eff} > 2$

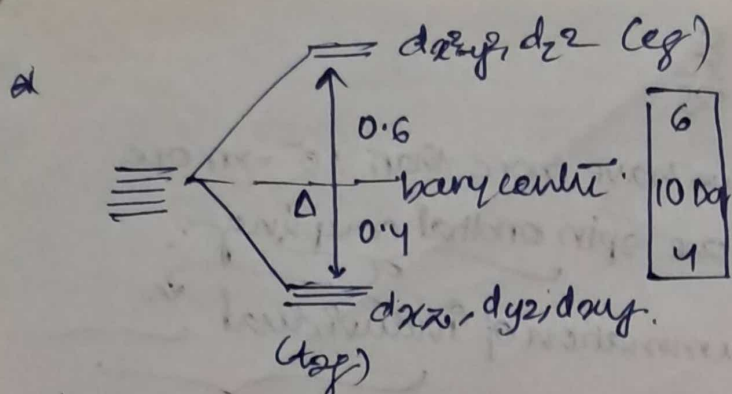
$$\lambda = \pm \frac{E}{2S}$$

when we have more than $1e^- \Rightarrow$ more than one spin orbital coupling.

$\Delta \Rightarrow$ separation (energy levels)
 (tagged eg)

$\Rightarrow$ summation of individual ϵ_i

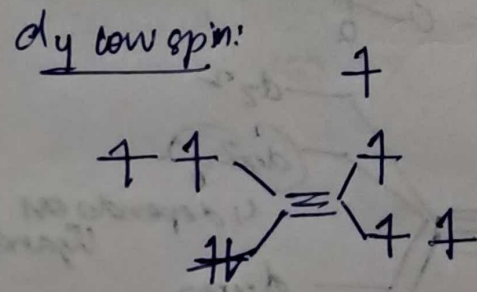
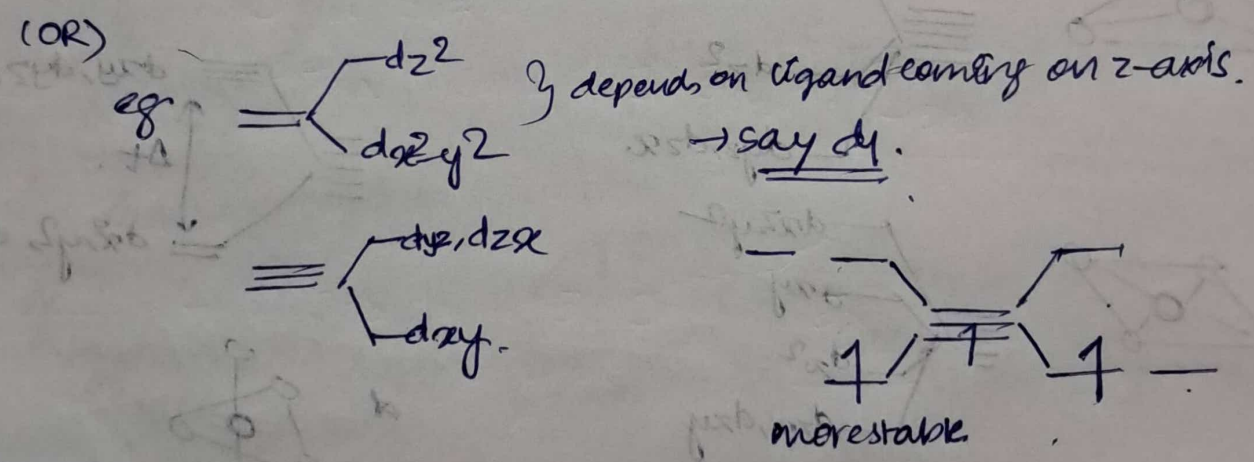
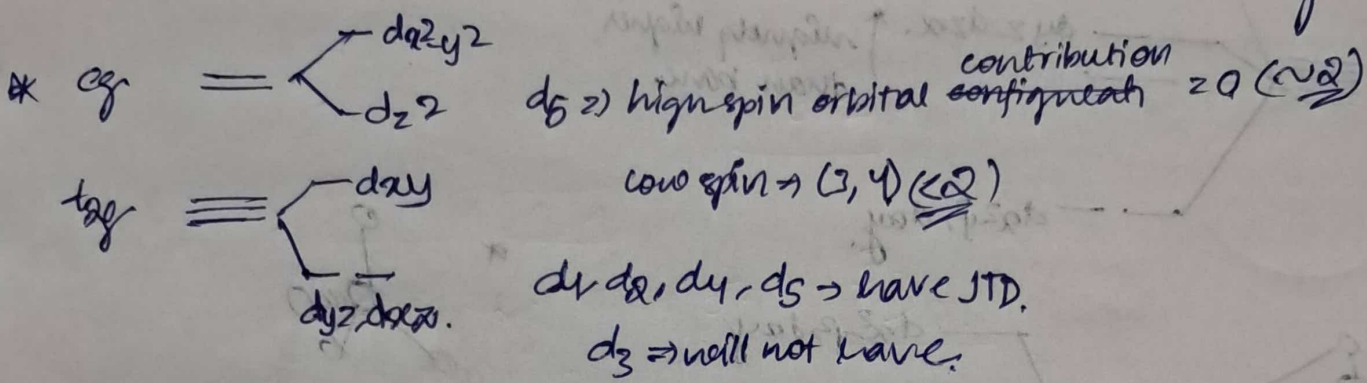




deg.
 (depends on ligand strength, pairing energy, & depends on size of orbitals)

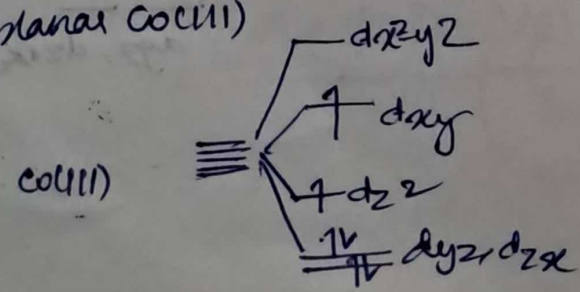
John-Teller distortion:

* Any non-linear molecular system in degenerated electronic state will be unstable. It will lift its degeneracy to get stability.

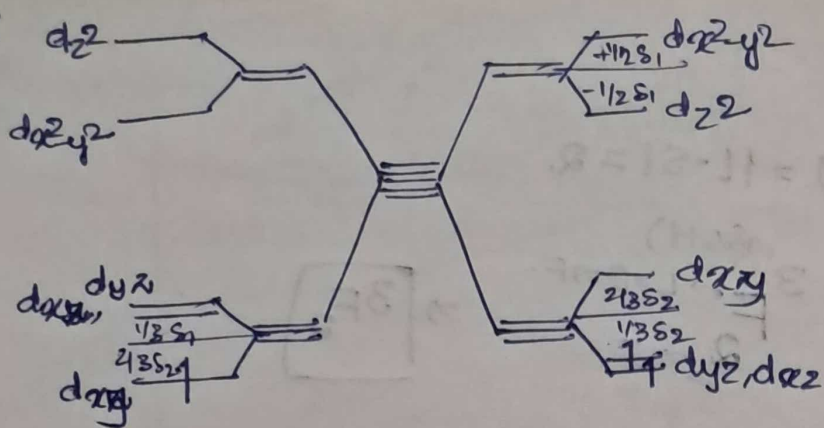


the spin orbital coupling constant $\propto$ degeneracy.

Ex square planar CoCl2



only CoCl2 is high spin, rest are low spin.
 could octahedral low spin.
 no of unpaired e- 2, 0, 4



$d_1 \Rightarrow d_{z^2}$ higher energy
 $\Rightarrow$ interaction is more along the z dirn
 $\Rightarrow$ 'Axially compressed' system.

$d_2 \Rightarrow d_{x^2-y^2}$ shorter bond length
 'Axially elongated' system.

$d_3 \Rightarrow$ no difference.

$d_4 \Rightarrow$ depends on the ligand.

$\hookrightarrow$ low spin: left more stable.

d_5 low spin: right.

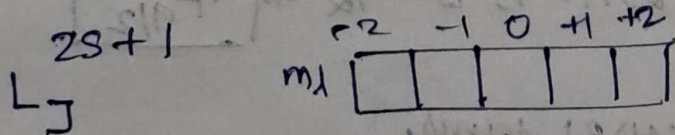
$\Rightarrow$ for tetrahedral, JTD is minimum 'cause the interaction is not head on with the orbitals $\Rightarrow$ splitting will not be as effective as in octahedral.

$\Rightarrow$ tetragonal CFSE < octahedral CFSE

$\hookrightarrow$ energy much lower $\Rightarrow$ more stable. ; $\delta_1 > \delta_2$ always.

(because it's a directional bond).

A Russell-Saunders Term Symbol:



$$L = \sum m_l ; L = 2$$

$$2S+1 = 2 \times \frac{1}{2} + 1 = 2$$

$$S = \sum m_s ; S = \frac{1}{2}$$

L =	0	1	2	3	4
	S	P	D	F	

$$J = |L+S| \dots |L-S|$$

$>$ half-filled

$<$ half-filled.

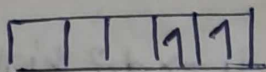
$$\Rightarrow 2D_{3/2}$$

$$\Rightarrow 2 - \frac{1}{2} = \frac{3}{2}$$

for d_4 system.

be clear with d-orbital at 4 in 10. you can 8 in be there na

d₂ system:



$$L = 2 + 1 = 3.$$

$$S = 2 \times \frac{1}{2} = 1$$

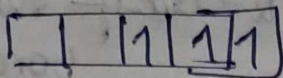
$$2S + 1 = 2 \times 1 + 1 = 3.$$

$$\Rightarrow J = |L - S| = 2.$$

$$\Rightarrow \overset{(2S+1)}{3} \overset{L=3}{F} \Rightarrow L=3 \Rightarrow F.$$

$$\Rightarrow \boxed{3F_2}$$

d₃ system:



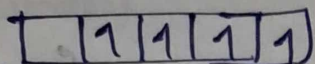
$$L = 3; S = 3/2.$$

$$2S + 1 = 4;$$

$$J = |L + S| = 9/2.$$

$$\Rightarrow \boxed{4F_{9/2}}$$

d₄ system:



$$L = 2; S = 2; 2S + 1 = 5.$$

$$J = |L - S| = 0$$

$$\Rightarrow \boxed{5D_0}$$

d₅ system:

$$L = 0; S = 5/2; 2S + 1 = 6; J = 5/2 \Rightarrow$$

$$\boxed{6S_{5/2}}$$

$$|L - S| \text{ or } |L + S|$$

doesn't matter as L=0 here

d₆ system:

$$L = 2; S = 2; 2S + 1 = 5; J = 0 \Rightarrow$$

$$\boxed{5D_0}$$

d₇ system:

$$L = 3; S = 3/2; 2S + 1 = 4; J = 9/2 \Rightarrow$$

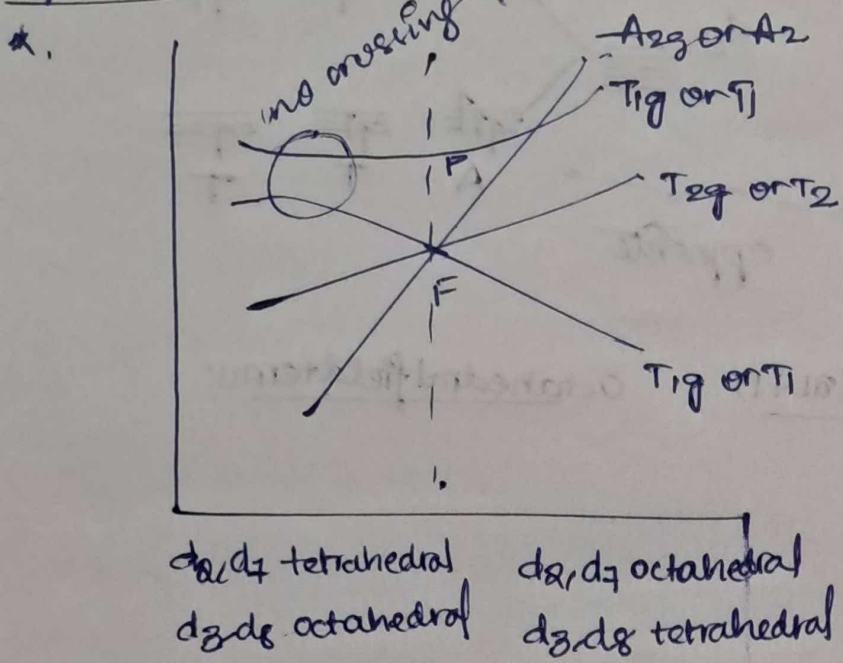
$$\boxed{4F_{9/2}}$$

$\Rightarrow d^n, d^{5+n}, d^{10-n} \Rightarrow$ same $(2S+1)$ & L terms.

$$D = \overset{2S+1}{F} E_g + \overset{2S+1}{T} E_g$$

$$F = \overset{2S+1}{T} E_g + \overset{2S+1}{T} E_g + \overset{2S+1}{A} E_g$$

Orbital diagrams



$d^n \Rightarrow d^{5+n}$ same

d^{5-n}
 d^{10-n} } opposite

(if they're of same symmetry
- they cannot cross each other.
↳ NO CROSSING RULE,
different symmetry
↳ can cross each other.

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$D \rightarrow T_{2g} E_g$ (octahedral)

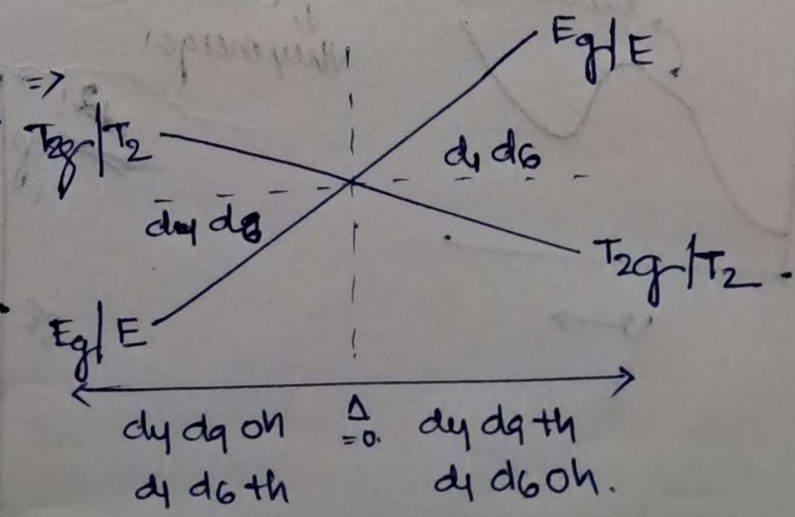
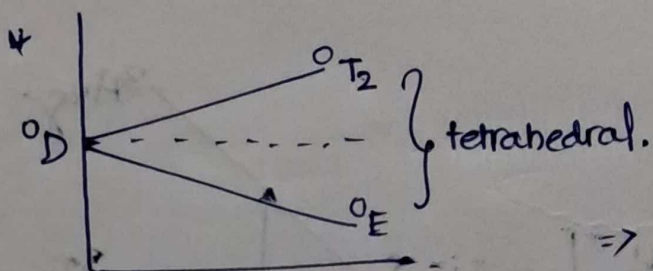
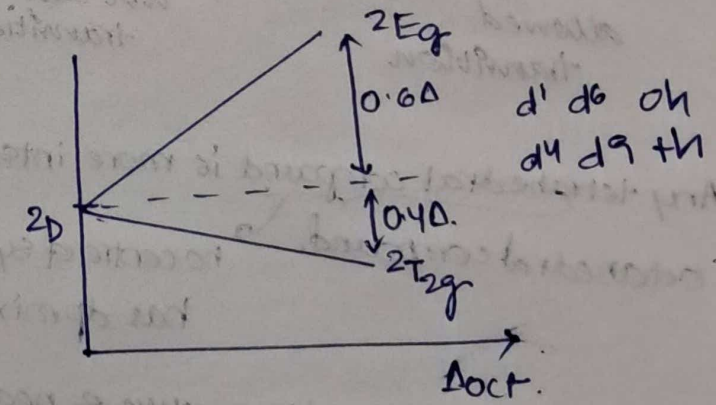
$T_2 E$ (tetrahedral)

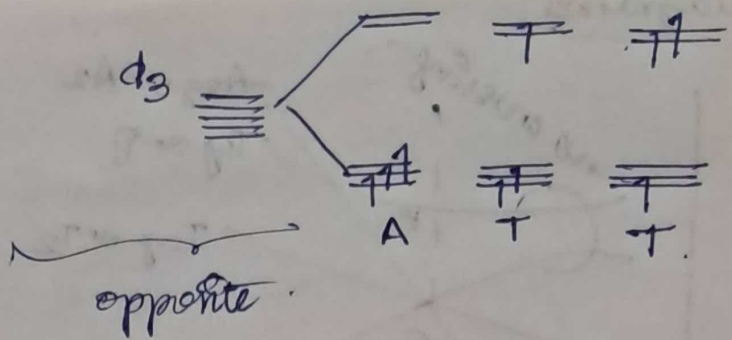
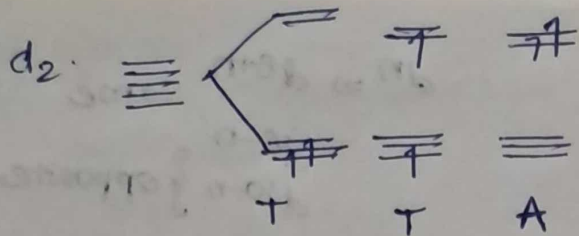
$F \rightarrow T_{1g} T_{2g} A_{2g}$

$T_1 T_2 A_2$

$d^n \leftrightarrow d^{5+n}$ same

$d^n \leftrightarrow d^{5-n} | d^{10-n}$ opposite.





* Free ion term:

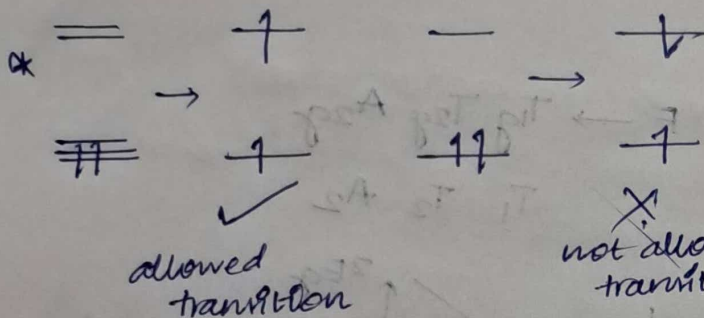
No. of states $(2L+1)$

Octahedral field terms:

$S(L=0)$

P

D



not allowed transition

Laporte

* $g \rightarrow g$ not allowed

$(p \rightarrow p)$

$g \rightarrow u$ allowed

* Any tetrahedral compound is more intensely coloured than its corresponding octahedral compound.

no centre of symmetry
has dpmixings

